MIL-C-10758D(MU)
2 September 1969
SUPERSEDING
MIL-C-10758C(CmlC)
11 April 1960
(See section 6)

MILITARY SPECIFICATION

CHEMICAL AGENT, GB

1. SCOPE

1.1 Scope. This specification covers two types of Chemical Agent, GB, chemically known as isopropyl methylphosphonofluoridate.

1.2 Classification. GB shall be of the following types as specified (see 6.2):

Type I - 81.0 minimum percent assay (for steel munitions).
Type II - Redistilled, 93.0 minimum percent assay (for aluminum or steel munitions).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-3250 - Cylinder, Gases and Liquids, One-Ton, Type D.
MIL-T-12014 - Tributylamine, Normal.
MIL-D-51161 - N,N'-Diisopropylcarbodiimide.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

FSC 1365

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DRAWINGS

US ARMY MUNITIONS COMMAND (EDGEWOOD ARSENAL)

D150-2-2 - Color Coding for Chemical Munitions.

TECHNICAL MANUALS

DEPARTMENT OF THE ARMY

TM 3-250 - Storage, Shipment, and Handling of Chemical Agents and Hazardous Chemicals.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publications should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Composition.

3.1.1 Type I. Type I GB shall consist of isopropyl methylphosphonofluoridate to which tributylamine conforming to MIL-T-12014 has been added as a stabilizer.

3.1.2 Type II. Type II GB shall consist of type I GB which has been redistilled and to which diisopropylcarbodiimide (DICDI) conforming to MIL-D-51161 has been added as an additional stabilizer.

3.2 Chemical characteristics. GB shall conform to the applicable chemical characteristics of table I when tested as specified therein.

Table I. Chemical characteristics

Characteristic	Type I	Type II	Test paragraph
Assay (as ispropyl methylphosphonofluoridate), as is basis after stabilization, minimum percent by weight	81.0	93.0	4.2.4.1
Acidity	To pass test	To pass test	4.2.4.2
Excess tributylamine, minimum percent by weight	1.0	----	4.2.4.3
Methylphosphonic difluoride, maximum percent by weight	----	0.50	4.2.4.4
Ionic fluoride, maximum percent by weight	----	0.20	4.2.4.5
Ionic chloride, maximum percent by weight	----	0.10	4.2.4.6
Excess DICDI, minimum percent by weight	----	2.0	4.2.4.7
Metals, maximum parts per million:			4.2.4.8
Copper	----	3.0	
Nickel	----	15.0	

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the GB of one type, which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the GB substantially uniform and which is contained in a single storage tank.

4.2.2 Sampling for test. Three representative specimens of between 50 and 60 milliliters (ml) each shall be randomly taken from each lot and placed in separate clean, dry containers labeled to identify the lot from which they were taken (see 6.3).

4.2.3 Inspection procedure. Each of the three specimens shall be separately tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. When applicable, blank determinations shall be run and corrections applied where significant. All tests shall be performed under a laboratory fume hood. Tests shall be conducted as follows (see 6.3):

4.2.4.1 Assay, as is basis after stabilization, (as isopropyl methylphosphonofluoridate).

(a) Indicator solution. Prepare a solution of 0.1 percent by weight of methyl red in methanol.

(b) Sodium hydroxide solution. Prepare a 0.1N carbonate-free aqueous sodium hydroxide solution and store in a polyethylene bottle. If this solution is to be stored for a long period of time, a tube containing a carbon dioxide absorbent should be attached.

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(c) Standard color solution. Prepare a standard color solution by adding 1 drop of 0.1N sodium hydroxide solution prepared as specified in (b) and 4 drops of indicator solution prepared as specified in (a) to 40 ml of water in a 200-ml Erlenmeyer flask.

(d) Procedure. Place 0.05 to 0.10 gram (g) of the specimen, weighed to the nearest 0.1 milligram (mg), in a glass bulb. Place the bulb in a 200-ml Erlenmeyer flask with 15 ml of water and 4 drops of indicator solution prepared as specified in (a). (For purposes of safety, it may be desirable to place the flask in another container, so that if the flask should break, the GB would be contained.) Swirl the flask gently to mix the indicator. Using a glass rod flattened on the end, crush the bulb and continue swirling the solution. To insure that the capillary is sufficiently broken, crush the tip first and work up to the bulb. Wash down the rod with 5 to 10 ml of water and titrate immediately and carefully with sodium hydroxide solution prepared as specified in (b) to a yellow end point while rapidly swirling the flask. The agitation of the flask should be made as efficiently as possible and the addition of the sodium hydroxide solution as uniformly as possible to avoid localized concentrations of sodium hydorxide solution. (The shorter the time for neutralization, the more likely it is that accurate results will be obtained.) Record the buret reading. Do not re-neutralize the solution if the yellow fades to pink after standing over 15 seconds. The precision of the initial end point can be determined by comparing the ratio of the milliliters of sodium hydroxide solution required for the first end point to the weight of the specimen in grams by duplicate determinations, or by the results from several analyses. This end point may be a major source of error if improperly titrated. Ratio values should be within $\pm$ 0.3 to obtain the desired precision. After obtaining the first end point, immediately and while swirling the solution, add 2.0 ml of the sodium hydroxide solution for each 0.01 g of specimen used. Record the buret reading and allow the flask to stand for 2 minutes. Back titrate with 0.1N hydrochloric acid to the first permanent change in the basic color (from yellow to orange) or until the first permanent change differing from the standard color solution prepared as specified in (c) is evident. Calculate the percent GB (as isopropyl methylphosphono-fluoridate) as follows:

$$\text{Percent GB} = \frac{7\,(AB - CD)}{W}$$

where: A = Milliliters of sodium hydroxide solution added after the initial end point,
B = Normality of the sodium hydroxide solution,
C = Milliliters of hydrochloric acid used,
D = Normality of hydrochloric acid, and
W = Weight of specimen in grams.

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4.2.4.2 Acidity. Place a bulb containing 0.3 to 0.4 g of the specimen in a flask with 30 ml of absolute ethanol and 1 ml of indicator solution prepared as specified in 4.2.4.3(a). Crush the bulb and swirl as specified in 4.2.4.1(d). The color of the solution shall remain green and shall not change to red.

4.2.4.3 Excess tributylamine.

(a) Indicator solution. Mix three parts of a 0.1-percent bromcresol green in absolute ethanol solution with one part of a 0.2-percent methyl red in absolute ethanol solution.

(b) Procedure. Place an ampoule containing approximately 1 g of the specimen weighed to the nearest 0.1 mg into a 300-ml Erlenmeyer flask containing 25 ml of absolute ethanol. Crush the ampoule with a tamping rod and wash the rod down with an additional 25 ml of absolute ethanol. Add several drops of indicator solution prepared as specified in (a) and titrate with 0.1N hydrochloric acid to the disappearance of the green color (the end point color is yellow to yellow-orange). Calculate the percent excess tributylamine as follows:

$$\text{Percent excess tributylamine} = \frac{18.53\ AB}{W}$$

where: A = Milliliters of hydrochloric acid used,
B = Normality of the hydrochloric acid, and
W = Weight of specimen in grams.

4.2.4.4 Methylphosphonic difluoride.

(a) Indicator solution. Mix three parts of a 0.1-percent bromcresol green in absolute ethanol solution, one part of a 0.2-percent methyl red in absolute ethanol solution, and 96 parts of absolute ethanol and allow to stand for 3 months.

(b) Procedure. Place a bulb containing 0.3 to 0.4 g of the specimen, weighed to the nearest 0.1 mg, in a 250-ml Erlenmeyer flask containing 10 ml of water, a liberal portion of crushed ice, and 10 to 15 drops of indicator solution prepared as specified in (a). Crush the bulb and swirl as specified in 4.2.4.1(d). Wash down the rod with 30 ml of water and immediately titrate, with rapid swirling, using 0.1N sodium hydroxide solution, to a deep green end point. Calculate the percent methylphosphonic difluoride as follows:

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$$\text{Percent methylphosphonic difluoride} = \frac{5.04\ AB}{W}$$

where: A = Milliliters of sodium hydroxide solution used,
B = Normality of sodium hydroxide solution, and
W = Weight of specimen in grams.

4.2.4.5 Ionic fluoride.

(a) Blank solution. Introduce 220 ml of absolute ethanol into a 500-ml round bottom flask with a 24/40 standard taper joint and connect a Liebig or similar condenser to the flask. Carefully add 10 g of sodium, divided into about 30 small pieces, through the top of the condenser, wait until the reaction is complete, and cool. (If reaction is too rapid, it may be slowed by placing the flask in an ice-water bath.) Transfer and rinse the solution into a 1-liter volumetric flask and dilute to the mark with water to make the blank solution.

(b) Thorium nitrate solution. Weigh 9.0 g of thorium nitrate [$Th(NO_3)_4 \cdot 4H_2O$] and dissolve in water to make 1 liter of solution (see 6.5).

(c) Buffer solution. Weigh 9.55 g of monochloroacetic acid and 2 g of sodium hydroxide and dissolve in water to make 100 ml of solution. Measure the pH of the solution by means of a pH meter and adjust the pH to 3.0 by adding monochloroacetic acid or sodium hydroxide, as applicable.

(d) Sodium fluoride solution. Weigh 2.2105 g of anhydrous sodium fluoride and dissolve in water to make 1 liter of solution.

(e) Standardization of thorium nitrate. Standardize the thorium nitrate against sodium fluoride and the other reagents, using an amount of fluoride approximately equivalent to the amount expected in the sample to be analyzed. Pipet 50-ml portions of the blank solution prepared as specified in (a) into each of ten 200-ml, tall-form beakers. Add varying amounts of sodium fluoride solution prepared as specified in (d) to each beaker (0.5, 1.0, 2.0, 3.0, 5.0, 7.0, 10.0, 15.0, 20.0, and 25.0 ml). Add a few drops of phenolphthalein to each beaker and make the solution just acid with 6N hydrochloric acid. Add 1 ml of 0.05-percent sodium alizarin sulfonate indicator. If the solution is pink, add more 6N hydrochloric acid until the color just disappears. Add 2-percent sodium hydroxide solution until there is a faint pink color. Add 0.05N hydrochloric acid until the pink color just disappears. Dilute to 100 ml with water. Add 4 ml of buffer solution prepared as specified in (c). (Check with a pH meter. This

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should buffer the solution to a pH of 3.1. If not, the buffer is not properly prepared. This is generally due to over-dilution in preparation.) Apply mechanical stirring and titrate the fluorine with thorium nitrate solution prepared as specified in (b) until a permanent pink color is obtained. With small amounts, a permanent pink color indicates the end-point; with larger amounts, a finely divided gelatinous precipitate is formed and the solution becomes slightly cloudy. The titration should be continued until the pink color is visible while stirring rapidly. Toward the end proceed slowly since the reaction between the flouride and thorium is slow. From the results obtained, plot a standardization curve, plotting grams of fluorine against milliliters of thorium nitrate.

(f) Procedure. Pipet 50 ml of blank solution prepared as specified in (a) into a tall-form beaker and add 2 drops of 1-percent phenolphthalein solution. Add 6N hydrochloric acid by drops until the solution is clear. Add 1 ml of 0.05-percent sodium alizarin sulfonate indicator and sufficient 6N hydrochloric acid to change the color of the solution to a pale yellow. Add 2-percent sodium hydroxide solution until the solution is faintly pink. Add 0.05N hydrochloric acid until the color of the solution changes from a faint pink to a pale yellow. Add 4 ml of buffer solution prepared as specified in (c). Weigh to the nearest 0.1 mg 0.3 to 0.7 g of the specimen into a glass bulb. Insert the bulbed specimen into the beaker and break with a glass rod. Wash down the rod with water and dilute to 100 ml. Check the pH of the solution with a standardized pH meter. If the pH is between 2.9 and 3.1, titrate with thorium nitrate solution prepared as specified in (b). If the pH is not between 2.9 and 3.1, discard the specimen and proceed as follows: To a blank solution that has been made just basic to sodium alizarin sulfonate indicator, add a new specimen and break the bulb with a glass rod. Adjust the pH with 2-percent alkali to faint pink of alizarin and with 0.05N acid to the first yellow. Add 4 ml of the buffer solution, dilute to 100 ml, and titrate immediately with thorium nitrate solution prepared as specified in (b) to prevent hydrolysis. (This procedure should only be necessary on highly acidic samples.) Read the number of grams of ionic fluorine directly from the titration curve prepared as specified in (e). Calculate the percent ionic fluoride as follows:

$$\text{Percent ionic fluoride} = \frac{100\ A}{W}$$

where: A = Weight of fluorine in grams from titration curve, and
W = Weight of specimen in grams.

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4.2.4.6 Ionic chlorine. Place a glass bulb containing 1.0 to 2.0 g of the specimen, weighed to the nearest milligram, in a 200-ml Erlenmeyer glass-stoppered flask containing 100 ml of 2-percent sodium hydroxide solution. Break the bulb by shaking the flask and allow the specimen to hydrolyze for 30 minutes. Acidify with nitric acid and determine the ionic chlorine content using the Volhard method, including filtration of the silver chloride precipitate prior to back titration with the standardized 0.1N potassium thiocyanate. Calculate the percent ionic chloride as follows:

$$\text{Percent ionic chloride} = \frac{3.546\ (AB - CD)}{W}$$

where: A = Milliliters of silver nitrate solution used,
B = Normality of silver nitrate solution,
C = Milliliters of potassium thiocyanate solution used,
D = Normality of potassium thiocyanate solution, and
W = Weight of specimen in grams.

4.2.4.7 Excess DICDI.

(a) Working curve. Using a syringe, place 1.2 ml of DICDI of known purity into a 100-ml, tared, glass-stoppered volumetric flask containing 50 ml of spectro grade carbon tetrachloride. Determine the weight of the added DICDI to the nearest 0.1 mg. Dilute to 100 ml with spectro grade carbon tetrachloride and mix thoroughly. (To avoid errors due to evaporation loss, the top 1/8 inch of the glass stopper should be greased with a silicone or other lubricant that is nonabsorbing at 4.75 microns.) Add aliquots of 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, and 10.0 ml of the DICDI solution to ten 50-ml volumetric flasks and dilute to 50 ml with spectro grade carbon tetrachloride. Insert glass stoppers, lubricated as specified above, into the volumetric flasks. Record the transmittance of each of the ten solutions by placing it in one cell of matched 0.50-millimeter cells and inserting it into an infrared spectrophotometer. Fill the second cell with spectro grade carbon tetrachloride and insert this cell into the reference beam of the spectrophotometer and scan between 4.5 and 5.0 microns. (The percent transmittance is set at 95, at 4.5 microns before starting the scan.) Plot a working curve of milligrams of pure DICDI per 50 ml of solution versus percent transmittance on semi-log paper.

(b) Procedure. Introduce 1 to 1.5 ml of specimen into a tared, glass-stoppered, 50-ml volumetric flask containing 20 ml of spectro grade carbon tetrachloride and determine the weight of the specimen to the nearest 0.1 mg. Dilute to 50 ml with spectro grade

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carbon tetrachloride, and mix. Fill a 0.50-millimeter cell with the specimen solution and a matched reference cell with spectro grade carbon tetrachloride and insert both into the spectrophotometer. With the wave length set for 4.5 microns and the cells at temperature, set the percent transmittance to 95 using the 100 percent adjust control. Scan between 4.5 and 5.0 microns. From the percent transmittance value, read the milligrams of DICDI present in the 50 ml of specimen solution from the working curve. Calculate the percent DICDI as follows:

$$\text{Percent DICDI} = \frac{A}{10\ W}$$

where: A = Milligrams of DICDI read from curve, and
W = Weight of specimen in grams.

4.2.4.8 Metals.

(a) Standard solutions.

(1) Copper. Dry approximately 0.5 g of National Bureau of Standards (NBS) Standard Sample No. 1056a, cupric cyclohexanebutyrate (16.1 percent copper) over fresh phosphorus pentoxide in a desiccator for 2 hours. Weigh 0.310 g of the dried standard into a 100-ml volumetric flask. Add 2 ml of xylene and 4 ml of 2-ethylhexylamine and heat on a hot plate until the solution is clear. Cool, and dilute to 100 ml with isopropyl alcohol. This solution contains 500 parts per million (ppm) of copper. Prepare three standard copper solutions containing 2, 5, and 10 ppm of copper by diluting the 500-ppm solution with appropriate amounts of isopropyl alcohol.

(2) Nickel. Dry approximately 0.35 g of NBS Standard Sample No. 1065a, nickel cyclohexanebutyrate (16.8 percent nickel) over fresh phosphorus pentoxide in a desiccator for 48 hours. Weigh 0.298 g of the dried standard into a 100-ml volumetric flask. Add 3 ml of xylene and 5 ml of 2-ethylhexanoic acid and heat on a hot plate until the solution is clear. Cool, and dilute to 100 ml with isopropyl alcohol. This solution contains 500 ppm of nickel. Prepare three standard nickel solutions containing 5, 10, and 15 ppm of nickel by diluting the 500-ppm solution with appropriate amounts of isopropyl alcohol.

(b) Preparation of working curves. Prepare a working curve for each metal as follows: Aspirate each standard solution into a Perkin-Elmer "Model 303" Atomic Absorption spectrophotometer or other suitable spectrophotometer. Operating conditions for the "Model 303" spectrophotometer

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are listed in table II. Conditions for other spectrophotometers will depend upon the instrument used. Determine the percent absorption of each standard solution and convert this reading to absorbance by use of table III. Check each standard on the proper correction line and if a correction is needed, convert it from percent absorption to absorbance. Subtract the correction absorbance from the absorbance of the standard solution to obtain the corrected total absorbance. Plot the corrected total absorbance against the respective standard solution concentration. This working curve should be reproducible at any later date with the same instrument and standard conditions. (Standards and working curves expressed as ppm are actually milligrams per liter of isopropyl alcohol. The final calculation using the dilution factor and density of GB expresses the results in true ppm which is equal to milligrams of metal per 1000 g of GB.)

Table II. Operating conditions for Perkin-Elmer "Model 303" Atomic Absorption Spectrophotometer

Operating condition	Copper	Nickel
Range	UV	UV
Wavelength	3247A	2320A
Slit	4	3
Source	15 ma	25 ma
Fuel (oxidizing flame)	Acetylene	Acetylene
Oxidizer	Air	Air
Correction wavelength	2961	----
Correction line	----	2317A
Zero at 25 percent A	----	----

(c) Procedure. Measure into a 10-ml volumetric flask, partially filled with isopropyl alcohol, 1.0 ml of specimen. Dilute to 10 ml with isopropyl alcohol and mix well. Aspirate the specimen solution into the spectrophotomer, which must be in a hood, using the operating conditions applicable to the metal being determined. Record the percent absorption and convert to absorbance by use of the conversion figures in table III. Using the correction line (where necessary), record the percent absorption and convert to absorbance. Subtract the corrected absorbance from the specimen solution absorbance to obtain the corrected total absorbance. Convert the corrected total absorbance to concentration by use of the applicable standard working curve. Multiply the concentration by the dilution factor (10) and the density of GB (0.926) to get the actual concentration of the metal in GB.

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Table III. Conversion of percent absorption to absorbance

To convert percent absorption (A) to absorbance, find the percent absorption to the nearest whole digit in the left-hand column; read across to the column located under the tenth of a percent desired and read the absorbance value. The value of absorbance corresponding to 26.8 percent is thus 0.1355.

A	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0.0	.0000	.0004	.0009	.0013	.0017	.0022	.0026	.0031	.0035	.0039
1.0	.0044	.0048	.0052	.0057	.0061	.0066	.0070	.0074	.0079	.0083
2.0	.0088	.0092	.0097	.0101	.0106	.0110	.0114	.0119	.0123	.0128
3.0	.0132	.0137	.0141	.0146	.0150	.0155	.0159	.0164	.0168	.0173
4.0	.0177	.0182	.0186	.0191	.0195	.0200	.0205	.0209	.0214	.0218
5.0	.0223	.0227	.0232	.0236	.0241	.0246	.0250	.0255	.0259	.0264
6.0	.0269	.0273	.0278	.0283	.0287	.0292	.0297	.0301	.0306	.0311
7.0	.0315	.0320	.0325	.0329	.0334	.0339	.0343	.0348	.0353	.0357
8.0	.0362	.0367	.0372	.0376	.0381	.0386	.0391	.0395	.0400	.0405
9.0	.0410	.0414	.0419	.0424	.0429	.0434	.0438	.0443	.0448	.0453
10.0	.0458	.0462	.0467	.0472	.0477	.0482	.0487	.0491	.0496	.0501
11.0	.0506	.0511	.0516	.0521	.0526	.0531	.0535	.0540	.0545	.0550
12.0	.0555	.0560	.0565	.0570	.0575	.0580	.0585	.0590	.0595	.0600
13.0	.0605	.0610	.0615	.0620	.0625	.0630	.0635	.0640	.0645	.0650
14.0	.0655	.0660	.0665	.0670	.0675	.0680	.0685	.0691	.0696	.0701
15.0	.0706	.0711	.0716	.0721	.0726	.0731	.0737	.0742	.0747	.0752
16.0	.0757	.0762	.0768	.0773	.0778	.0783	.0788	.0794	.0799	.0804
17.0	.0809	.0814	.0820	.0825	.0830	.0835	.0841	.0846	.0851	.0857
18.0	.0862	.0867	.0872	.0878	.0883	.0888	.0894	.0899	.0904	.0910
19.0	.0915	.0921	.0926	.0931	.0937	.0942	.0947	.0953	.0958	.0964
20.0	.0969	.0975	.0980	.0985	.0991	.0996	.1002	.1007	.1013	.1018
21.0	.1024	.1029	.1035	.1040	.1046	.1051	.1057	.1062	.1068	.1073
22.0	.1079	.1085	.1090	.1096	.1101	.1107	.1113	.1118	.1124	.1129
23.0	.1135	.1141	.1146	.1152	.1158	.1163	.1169	.1175	.1180	.1186
24.0	.1192	.1198	.1203	.1209	.1215	.1221	.1226	.1232	.1238	.1244
25.0	.1249	.1255	.1261	.1267	.1273	.1278	.1284	.1290	.1296	.1302
26.0	.1308	.1314	.1319	.1325	.1331	.1337	.1343	.1349	.1355	.1361
27.0	.1367	.1373	.1379	.1385	.1391	.1397	.1403	.1409	.1415	.1421
28.0	.1427	.1433	.1439	.1445	.1451	.1457	.1463	.1469	.1475	.1481
29.0	.1487	.1494	.1500	.1506	.1512	.1518	.1524	.1530	.1537	.1543

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Table III. Conversion of percent absorption to absorbance (continued)

A	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
30.0	.1549	.1555	.1561	.1568	.1574	.1580	.1586	.1593	.1599	.1605
31.0	.1612	.1618	.1624	.1630	.1637	.1643	.1649	.1656	.1662	.1669
32.0	.1675	.1681	.1688	.1694	.1701	.1707	.1713	.1720	.1726	.1733
33.0	.1739	.1746	.1752	.1759	.1765	.1772	.1778	.1785	.1791	.1798
34.0	.1805	.1811	.1818	.1824	.1831	.1838	.1844	.1851	.1858	.1864
35.0	.1871	.1878	.1884	.1891	.1898	.1904	.1911	.1918	.1925	.1931
36.0	.1938	.1945	.1952	.1959	.1965	.1972	.1979	.1986	.1993	.2000
37.0	.2007	.2013	.2020	.2027	.2034	.2041	.2048	.2055	.2062	.2069
38.0	.2076	.2083	.2090	.2097	.2104	.2111	.2118	.2125	.2132	.2140
39.0	.2147	.2154	.2161	.2168	.2175	.2182	.2190	.2197	.2204	.2211
40.0	.2218	.2226	.2233	.2240	.2248	.2255	.2262	.2269	.2277	.2284
41.0	.2291	.2299	.2306	.2314	.2321	.2328	.2336	.2343	.2351	.2358
42.0	.2366	.2373	.2381	.2388	.2396	.2403	.2411	.2418	.2426	.2434
43.0	.2441	.2449	.2457	.2464	.2472	.2480	.2487	.2495	.2503	.2510
44.0	.2518	.2526	.2534	.2541	.2549	.2557	.2565	.2573	.2581	.2588
45.0	.2596	.2604	.2612	.2620	.2628	.2636	.2644	.2652	.2660	.2668
46.0	.2676	.2684	.2692	.2700	.2708	.2716	.2725	.2733	.2741	.2749
47.0	.2757	.2765	.2774	.2782	.2790	.2798	.2807	.2815	.2823	.2832
48.0	.2840	.2848	.2857	.2865	.2874	.2882	.2890	.2899	.2907	.2916
49.0	.2924	.2933	.2941	.2950	.2958	.2967	.2976	.2984	.2993	.3002
50.0	.3010	.3019	.3028	.3036	.3045	.3054	.3063	.3072	.3080	.3089
51.0	.3098	.3107	.3116	.3125	.3134	.3143	.3152	.3161	.3170	.3179
52.0	.1388	.3197	.3206	.3215	.3224	.3233	.3242	.3251	.3261	.3270
53.0	.3279	.3288	.3298	.3307	.3316	.3325	.3335	.3344	.3354	.3363
54.0	.3372	.3382	.3391	.3401	.3410	.3420	.3429	.3439	.3449	.3458
55.0	.3468	.3478	.3487	.3497	.3507	.3516	.3526	.3536	.3546	.3556
56.0	.3565	.3575	.3585	.3595	.3605	.3615	.3625	.3635	.3645	.3655
57.0	.3665	.3675	.3686	.3696	.3706	.3716	.3726	.3737	.3747	.3757
58.0	.3768	.3778	.3788	.3799	.3809	.3820	.3830	.3840	.3851	.3862
59.0	.3872	.3883	.3893	.3904	.3915	.3925	.3936	.3947	.3958	.3969
60.0	.3979	.3990	.4001	.4012	.4023	.4034	.4045	.4056	.4067	.4078
61.0	.4089	.4101	.4112	.4123	.4134	.4145	.4157	.4168	.4179	.4191
62.0	.4202	.4214	.4225	.4237	.4248	.4260	.4271	.4283	.4295	.4306
63.0	.4318	.4330	.4342	.4353	.4365	.4377	.4389	.4401	.4413	.4425
64.0	.4437	.4449	.4461	.4473	.4485	.4498	.4510	.4522	.4535	.4547
65.0	.4559	.4572	.4584	.4597	.4609	.4622	.4634	.4647	.4660	.4672

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Table III. Conversion of percent absorption to absorbance (continued)

A	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
66.0	.4685	.4698	.4711	.4724	.4737	.4750	.4763	.4776	.4789	.4802
67.0	.4815	.4828	.4841	.4855	.4868	.4881	.4895	.4908	.4921	.4935
68.0	.4948	.4962	.4976	.4989	.5003	.5017	.5031	.5045	.5058	.5072
69.0	.5086	.5100	.5114	.5129	.5143	.5157	.5171	.5186	.5200	.5214
70.0	.5229	.5243	.5258	.5272	.5287	.5302	.5317	.5331	.5346	.5361
71.0	.5376	.5391	.5406	.5421	.5436	.5452	.5467	.5482	.5498	.5513
72.0	.5528	.5544	.5560	.5575	.5591	.5607	.5622	.5638	.5654	.5670
73.0	.5686	.5702	.5719	.5735	.5751	.5768	.5784	.5800	.5817	.5834
74.0	.5850	.5867	.5884	.5901	.5918	.5935	.5952	.5969	.5986	.6003
75.0	.6021	.6038	.6055	.6073	.6091	.6108	.6126	.6144	.6162	.6180
76.0	.6198	.6216	.6234	.6253	.6271	.6289	.6308	.6326	.6345	.6364
77.0	.6383	.6402	.6421	.6440	.6459	.6478	.6498	.6517	.6536	.6556
78.0	.6576	.6596	.6615	.6635	.6655	.6676	.6696	.6716	.6737	.6757
79.0	.6778	.6799	.6819	.6840	.6861	.6882	.6904	.6925	.6946	.6968
80.0	.6990	.7011	.7033	.7055	.7077	.7100	.7122	.7144	.7167	.7190
81.0	.7212	.7235	.7258	.7282	.7305	.7328	.7352	.7375	.7399	.7423
82.0	.7447	.7471	.7496	.7520	.7545	.7570	.7595	.7620	.7645	.7670
83.0	.7696	.7721	.7747	.7773	.7799	.7825	.7852	.7878	.7905	.7932
84.0	.7959	.7986	.8013	.8041	.8069	.8097	.8125	.8153	.8182	.8210
85.0	.8239	.8268	.8297	.8327	.8356	.8386	.8416	.8447	.8477	.8508
86.0	.8539	.8570	.8601	.8633	.8665	.8697	.8729	.8761	.8794	.8827
87.0	.8861	.8894	.8928	.8962	.8996	.9031	.9066	.9101	.9136	.9172
88.0	.9208	.9245	.9281	.9318	.9355	.9393	.9431	.9469	.9508	.9547
89.0	.9586	.9626	.9666	.9706	.9747	.9788	.9830	.9872	.9914	.9957

5. PREPARATION FOR DELIVERY

This section is applicable to type I GB only.

5.1 Packing, level A. GB shall be packed in accordance with Department of Transportation (DOT) regulations in either new or reconditioned type D, one-ton cylinders conforming to MIL-C-3250. If reconditioned cylinders are used, each cylinder shall have been reconditioned in accordance with procedures of TM 3-250 or as otherwise specified (see 6.2).

5.2 Marking. In addition to any marking required by the contract or order, cylinders shall be marked in accordance with DOT regulations, Drawing D150-2-2, and MIL-STD-129. DA label 69, "Warning, Beware of Fumes - Poison Gas," shall be applied to each cylinder (see 6.2).

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6. NOTES

6.1 Intended use. Type I GB is intended for use in filling steel munitions. Type II GB is intended for use in filling aluminum or steel munitions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of GB required (see 1.2).
(c) Reconditioning procedure other than that specified, if required (see 5.1).
(d) Additional marking, if required (see 5.2).

6.3 Caution. The material covered by this specification is extremely toxic and unless special precautions are taken by personnel using it or handling it, fatalities will result. Safety information is available in AMCR 385-233 for the processing, filling, handling, and decontamination of GB.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Radiation precaution. If the levels of thorium nitrate as specified in Title 10, CFR, part 40 are exceeded (15 pounds) on hand at any one time, then the provisions of Title 10, CFR, part 40 should be followed.

6.6 Supersession data. Type I GB is the material formerly covered by MIL-C-10758C(CmlC). Type II GB includes the requirements of Edgewood Arsenal Purchase Description 196-21-86, dated 14 June 1968.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1365-A038

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC–WPAFB–OCT 67 2M

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